10

four different chemical languages

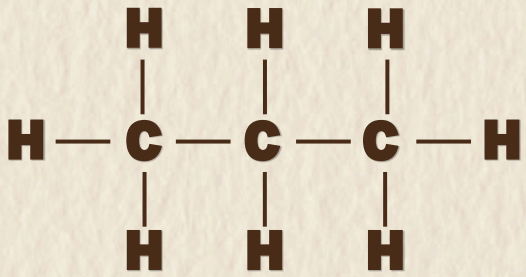
PROPANE

Propane is a liquefied petroleum gas that comes out of both oil and gas wells.

MOLECULAR FORMULA

C_3H_8

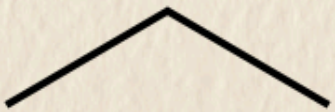
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

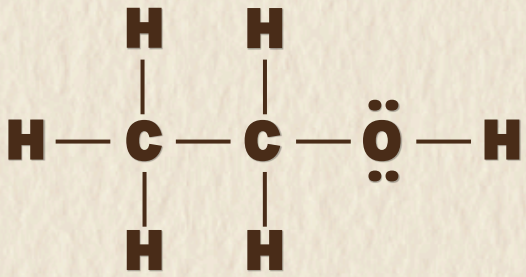
ETHANOL

Ethanol, also known as ethyl alcohol or simply alcohol, is a chemical compound that is both colorless and flammable.

MOLECULAR FORMULA

C_2H_6O

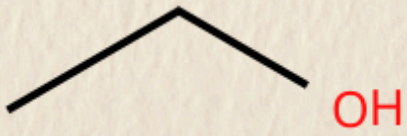
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

CH_3CH_2OH

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

propan-2-one

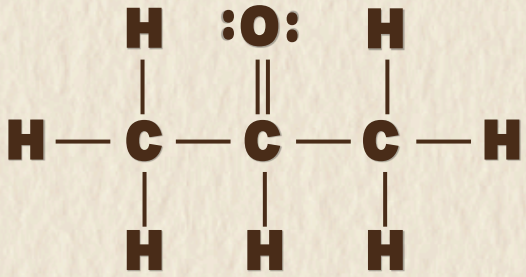
ACETONE

Acetone is a liquid solvent that can break down and dissolve other substances.

MOLECULAR FORMULA

C_3H_6O

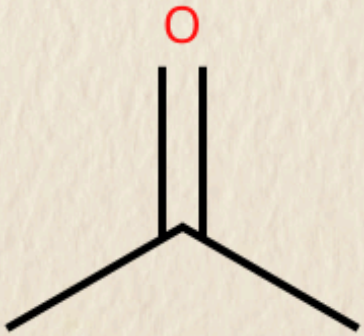
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

CH_3COCH_3

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

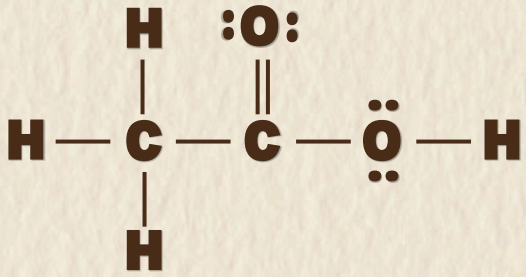
ACETIC ACID

Acetic acid, also known as ethanoic acid, is a chemical compound found in many different products. It's perhaps most well-known as the main component of vinegar.

MOLECULAR FORMULA

$C_2H_4O_2$

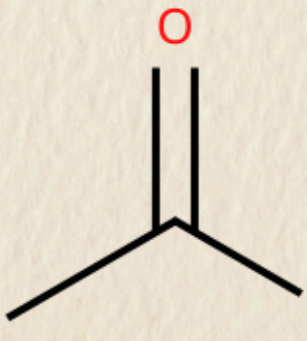
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

CH_3COOH

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

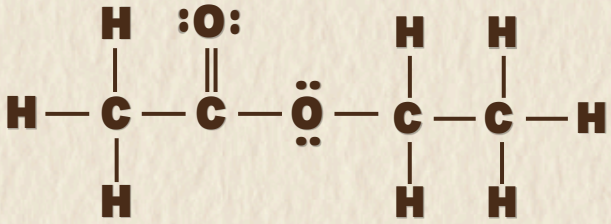
ETHYL ACETATE

Ethyl Acetate is an organic ester compound that appears as a colorless liquid. It is highly miscible with all common organic solvents.

MOLECULAR FORMULA

$C_4H_8O_2$

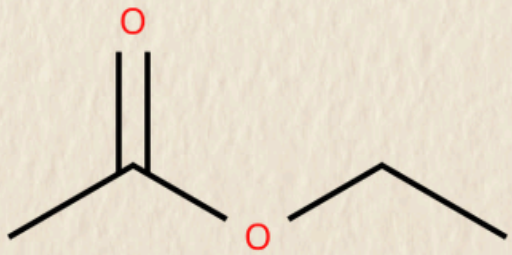
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3COOC_2H_5$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

90

four different chemical languages

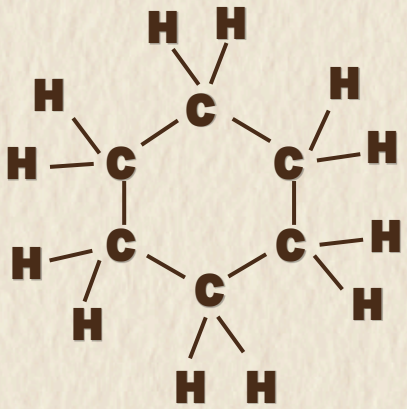
CYCLOHEXANE

Cyclohexane appears as a clear colorless liquid with a petroleum-like odor. Used to make nylon, as a solvent, paint remover, and to make other chemicals.

MOLECULAR FORMULA

C_6H_{12}

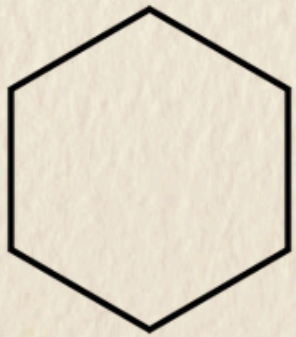
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$(CH_2)_6$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

07

four different chemical languages

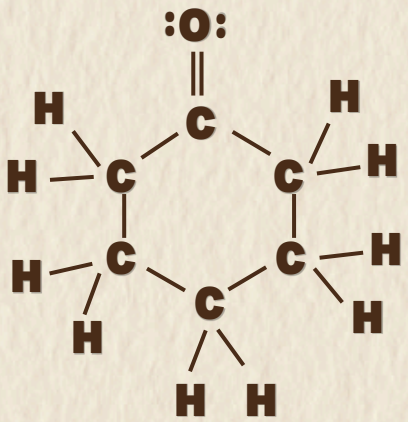
CYCLOHEXANONE

Cyclohexanone is an organic compound that is a colorless to pale yellow liquid with an odor similar to that of acetone that is very soluble in most organic solvents.

MOLECULAR FORMULA

$C_6H_{10}O$

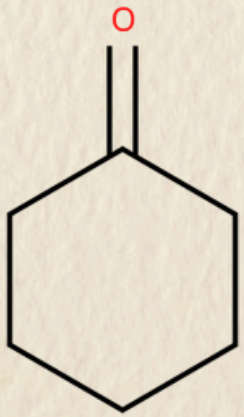
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$(CH_2)_5CO$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

N-(4-hydroxyphenyl)acetamide

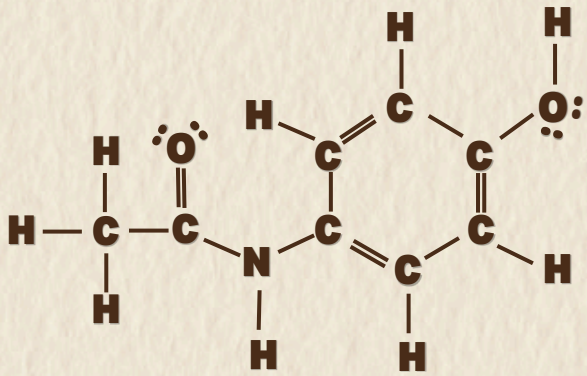
ACETAMINOPHEN

Acetaminophen (paracetamol), also commonly known as Tylenol, is the most commonly taken analgesic worldwide and is recommended as first-line therapy in pain conditions by the World Health Organization (WHO).

MOLECULAR FORMULA

$C_8H_9NO_2$

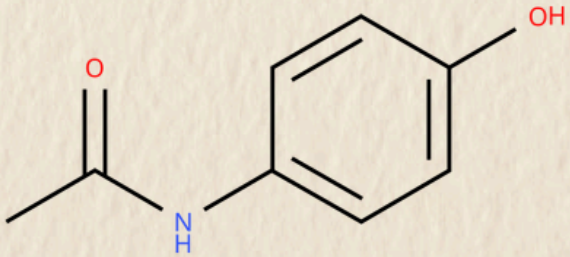
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$HOC_6H_4NHCOCH_3$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

2-acetyloxybenzoic acid

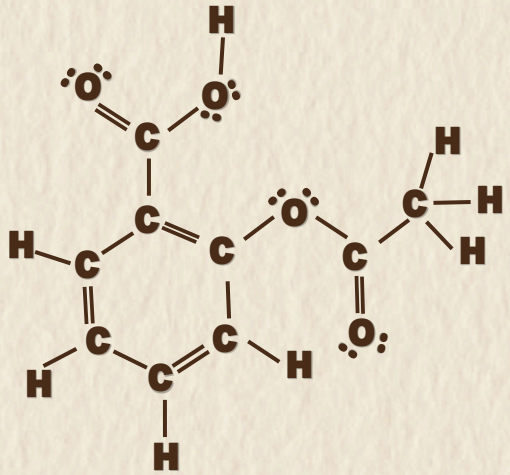
ASPIRIN

Aspirin (acetylsalicylic acid) is one of the oldest and most commonly used non-steroidal anti-inflammatory drugs (NSAIDs) for treating mild-to-moderate pain, fever, and inflammation from various causes.

MOLECULAR FORMULA

$C_9H_8O_4$

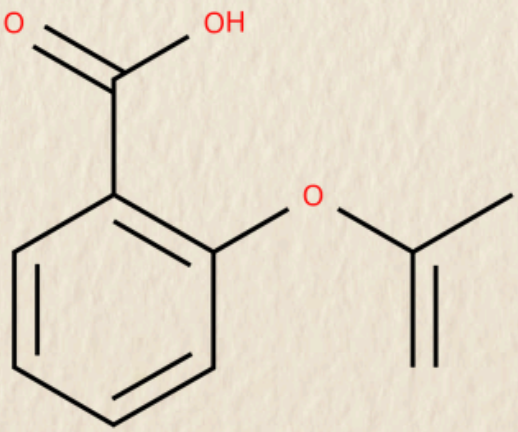
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3COOC_6H_4COOH$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name

four different chemical languages

1,3,7-trimethylpurine-2,6-dione

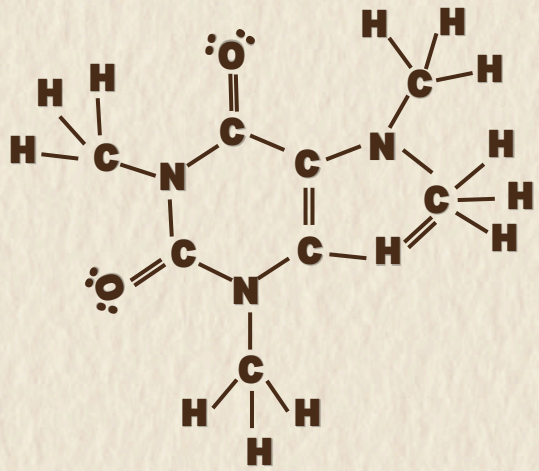
CAFFEINE

Caffeine is a drug of the methylxanthine class used for a variety of purposes, including certain respiratory conditions of the premature newborn, pain relief, and to combat drowsiness.

MOLECULAR FORMULA

$C_8H_{10}N_4O_2$

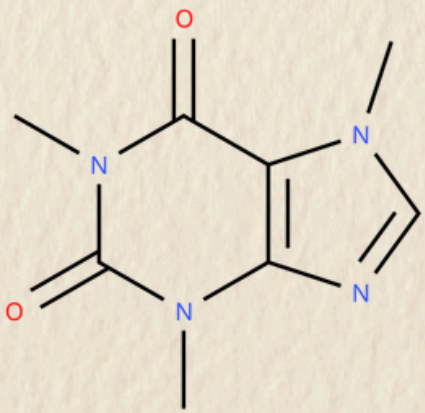
LEWIS STRUCTRE



CONDENSED STRUCTURAL FORMULA

$CH_3N(CH_3)C(=O)C_1=C(NC=N_1)N(C)C(=O)$

BOND-LINE FORMULA



- [x] atom count
- [x] skeletal notation
- [x] connectivity consistent
- [x] valence
- [x] IUPAC name